

For each question, show all related work and explain the solution. Giving only a correct solution with no or incomplete work, no explanation and/or improper notation can result in a score of 0 for that question.

1. 5 pts each - Suppose a "word" is any 7 letters of the alphabet with repeats allowed. Find the number of "words" with the following characteristics. Be sure to show your calculation and solution for each.

- a. The total number of "words" (no restrictions)

There are 26 choices for 7 independent positions.

$$26 \times 26 \times 26 \times 26 \times 26 \times 26 \times 26 = 26^7 = 8,031,810,176$$

- b. How many "words" begin with the letter "A"

26 choices for 6 independent positions, 1 is fixed at "A"

$$1 \times 26 \times 26 \times 26 \times 26 \times 26 \times 26 = 26^6 = 308,915,776$$

- c. How many "words" begin with "S" and end with "F"

26 choices for 5 independent positions, 2 are fixed.

$$1 \times 26 \times 26 \times 26 \times 26 \times 26 \times 1 = 26^5 = 11,881,376$$

- d. How many "words" begin with "ABC" in some order?

26 choices for 4 independent positions, first 3 are a permutation of {A, B, C}

$$3! \times 26 \times 26 \times 26 \times 26 = 6 \times 26^4 = 2,741,856$$

- e. How many "words" have exactly one vowel (A, E, I, O, U)?

Position of vowel: $\binom{7}{1} = \frac{7!}{1!6!} = 7$ ways

5 choices for that vowel

$$7 \times 5 \times 21^6 = 35 \times 21^6 = 3,001,814,235$$

Rest are 21 choices for 6 independent positions.

2. 10 pts - The given solution is incorrect. Find the error, explain why it is incorrect and how you would correctly solve the problem. The correct solution should relate to the incorrect solution. Show any calculations necessary.

Question: Find the number of ways to get two pairs of two different types of cards (like 2 3's and 2 7's) in a hand of 4 cards from a standard 52-card deck.

Incorrect Solution: There are 13 choices for the type of card for the first pair and $\binom{4}{2}$ ways to choose the cards in that pair. There are 12 choices left for the type of card in the second pair and $\binom{4}{2}$ ways to choose the cards in that pair. Therefore, the total number of ways to get two pair is:

$$13 \cdot \binom{4}{2} \cdot 12 \cdot \binom{4}{2}$$

Error: This solution counts each two pair hand twice. A hand is counted once when, let's say, first pair = 3's and second pair = 7's, and is counted again when first pair = 7's and second pair = 3's. But order does not matter in card hands, so we must divide by 2!

$$\frac{13 \cdot 12}{2} \cdot \binom{4}{2} \cdot \binom{4}{2} = 78 \cdot 6^2 = 78 \cdot 36 = 2,808$$

3. 10 points - What is the probability that a positive integer less than 100 picked at random has all distinct digits?

Sample space: positive integers 1-99 = 99 outcomes.

1 digit numbers: all digits are distinct. 9 choices

2 digit numbers: tens digits = 9 choices

One digit = any 0-9 without matching tens digit = 9

$$\text{Total} = 9 + 81 = 90$$

$$\text{Probability} = \frac{90}{99} = \frac{10}{11} = 0.90909$$

4. 10 points - Suppose that two cards are drawn without replacement from a well-shuffled deck. What is the probability that both cards have numbers and that the numbers on the cards are the same (note that only the numbers 2 through 10 are shown on cards, since aces, kings, queens, and jacks are represented by letters).

$$\text{Total 2-card hands: } \binom{52}{2} = \frac{52!}{2!50!} = 1326$$

$$\text{Choose 1 of the numbered ranks: } \binom{9}{1} = \frac{9!}{1!8!} = 9$$

$$\text{Choose 2 suits of that rank: } \binom{4}{2} = \frac{4!}{2!2!} = 6$$

Probability:

$$\frac{\binom{9}{1} \binom{4}{2}}{\binom{52}{2}} = \frac{9 \cdot 6}{1326} = \frac{9}{221}$$

5. 10 points - A pair of dice numbered 1-6 is loaded so that on the first die, the probability of a 3 appearing is $\frac{3}{8}$, with each other outcome appearing with a probability of $\frac{1}{8}$. On the second die, the probability of a 4 appearing is $\frac{2}{7}$, with each other outcome having a probability of $\frac{1}{7}$. What is the probability that the sum of the numbers rolled is equal to 7?

The sum is 7 for the ordered pairs: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1)

The outcomes of the two dice are independent, so multiply the probabilities and add.

$$P(1,6) = \frac{1}{8} \cdot \frac{1}{7} = \frac{1}{56}$$

$$P(2,5) = \frac{1}{8} \cdot \frac{1}{7} = \frac{1}{56}$$

$$P(3,4) = \frac{3}{8} \cdot \frac{2}{7} = \frac{6}{56} = \frac{3}{28}$$

$$P(4,3) = \frac{1}{8} \cdot \frac{1}{7} = \frac{1}{56}$$

$$P(5,2) = \frac{1}{8} \cdot \frac{1}{7} = \frac{1}{56}$$

$$P(6,1) = \frac{1}{8} \cdot \frac{1}{7} = \frac{1}{56}$$

$$\frac{1}{56} \times 5 + \frac{3}{28} = \frac{5}{56} + \frac{6}{56} = \frac{11}{56} \approx 0.19643$$

6. 10 points - Suppose $|A| = |B| = |C| = 100$, $|A \cap B| = 60$, $|A \cap C| = 50$, $|B \cap C| = 40$, and $|A \cup B \cup C| = 175$. Find $|A \cap B \cap C|$.

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

$$175 = 100 + 100 + 100 - 60 - 50 - 40 + |A \cap B \cap C|$$

$$175 = 150 + |A \cap B \cap C|$$

$$|A \cap B \cap C| = 25$$

7. 15 points - Find a recurrence relation and initial condition for the number of fruit flies in a jar if there are 12 flies initially and every week there are six times as many flies in the jar as there were the previous week. Then find the explicit function.

Recurrence and initial condition

Let a_n be the number of flies n weeks after the start.

$$a_0 = 12$$

Initial Condition: $a_0 = 12$

$$a_1 = 12 \cdot 6 = 72$$

Each week there are six times as many flies as the previous week,

$$a_2 = 72 \cdot 6 = 432$$

$$a_n = 6a_{n-1} \text{ for } n \geq 1.$$

Explicit formula

This is a geometric sequence with a common ratio of 6. Thus,

$$a_n = a_0 \cdot 6^n = 12 \cdot 6^n$$

$$a_1 = 12 \cdot 6^1 = 72 \checkmark \quad a_2 = 12 \cdot 6^2 = 12 \cdot 36 = 432 \checkmark$$

8. 10 points - In a survey of 280 teenagers, it is found that 85 like carrots, 47 like peas, 74 like cauliflower, 22 like both carrots and peas, 32 like both peas and cauliflower, 42 like both carrots and cauliflower and only 12 like all 3 vegetables. How many teenagers don't like any of the vegetables?

Let C, P, F be the sets of teens who like carrots, peas, and cauliflower.

$$\text{Given: } |C| = 85, |P| = 47, |F| = 74$$

$$|C \cap P| = 22, |P \cap F| = 32, |C \cap F| = 42$$

$$|C \cap P \cap F| = 12$$

By inclusion-exclusion, the number who like at least one is:

$$|C \cup P \cup F| = |C| + |P| + |F| - |C \cap P| - |P \cap F| - |C \cap F| + |C \cap P \cap F|$$

$$= (85 + 47 + 74) - (22 + 32 + 42) + 12$$

$$= 122$$

$$\text{Total teens} = 280$$

Therefore, those who like none:

$$280 - 122 = 158$$